

Institute of Engineering & Technology, Indore
BE II-year Examination April-May 2024
Electronics and Telecommunication
Subject: Analog Communication
Subject Code: 4ETRC4

Time 3 Hour

Max. Marks: 60

Note: Attempt all questions and carry equal marks.**Q.1. a)** Expand the function $f(x)$ in Fourier series

$$\begin{aligned} f(x) &= \pi + x & -\pi < x < 0 \\ f(x) &= \pi - x & 0 < x < \pi \end{aligned}$$

and deduced

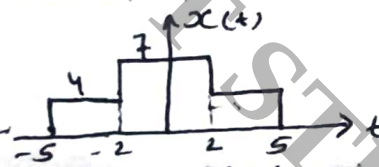
$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \dots$$

b) Express the function as

$$\begin{aligned} f(x) &= \sin x & 0 < x < \pi \\ f(x) &= 0 & x > \pi \end{aligned}$$

as sin integral. Evaluate $\int_0^\infty \frac{\sin 2\pi \sin \alpha x}{1-\alpha^2} d\alpha$ **c)** Find the Fourier transform of the given signals $f(x)$:

(i)



$$(ii) \text{Rect}_{(t-\frac{1}{2})} = 1 \quad -\frac{1}{2} < t < \frac{1}{2}$$

find the Fourier transform of $[x(t) + x(-t)]$ **Q.2. a)** An amplitude-modulated wave is represented by the expression

$$V(t) = 5[1 + 0.6\cos 6280t] \sin 2\pi \cdot 10^4 t$$

- (i) What are the minimum and maximum amplitudes of AM waves?
(ii) What frequency components are contained in the modulated wave and what is the amplitude of each component?

b) The antenna current of an AM transmitter is 8A when only the carrier is sent, but it increases to 8.93A when the carrier is sinusoidally modulated. Find the percentage of modulation

c) An amplitude-modulated signal is given by

$$V(t) = 0.1[1 + 0.1\cos 2512t + 0.5\cos 6280t] \sin(10^7 t + 45^\circ)$$

What information do you get from this signal and plot frequency spectrum?

Q.3. a) Define Narrowband FM (NBFM) and Wideband FM (WBFM) and describe their respective spectra. Explain the differences in transmission bandwidth between NBFM and WBFM. How does the modulation index influence the bandwidth in each case?

- b) Describe the direct method of Frequency Modulation generation, including the derivation of the expression for FM using this method 6
- c) Explain the principle of frequency demodulation using the simple slope method. Describe the steps involved in demodulating a frequency-modulated signal using this method 6
- Q.4. a) Explain the block diagram of an AM transmitter. Discuss the function of each block in the transmitter chain and how they contribute to the modulation process. Illustrate your answer with a labeled diagram. 6
- b) Describe the Tuned Radio Frequency (TRF) receiver architecture and its operation. Explain the function of each component in the TRF receiver circuit, including the RF amplifier, mixer, local oscillator, and detector. 6
- c) Explain the operation of a Frequency Modulation (FM) transmitter by detailing its block diagram. Provide a comprehensive overview of each block within the FM transmitter circuit, outlining the roles and functions of components such as the modulator, oscillator, power amplifier, and antenna coupling stage. 6
- Q.5. Write short notes on any 3 of the following: 12
- (i) Nyquist Sampling Theorem
 - (ii) Pulse Width Modulation
 - (iii) Pulse Amplitude Modulation
 - (iv) Pulse position Modulation

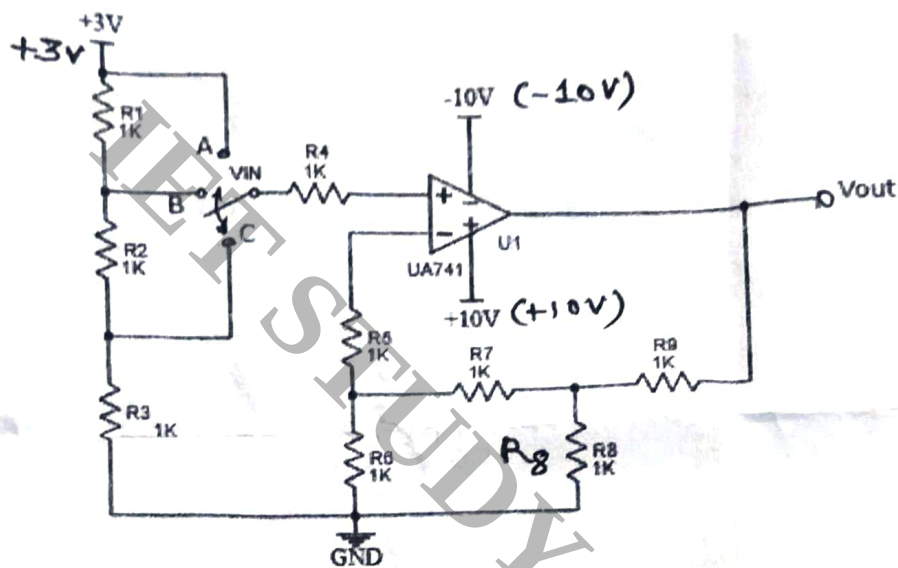
LINEAR DEVICES AND APPLICATIONS(4ETRC3)

B.E. 2nd YEAR - ELECTRONICS & TELECOMMUNICATIONS

Max marks:60
Time : 3 Hrs

Attempt all questions . Solve any two parts from each question. Do not use unfair means . Each question is of 12 Marks and each part is of 06 marks. Make suitable assumptions if required.

1. Consider the circuit shown in the following figure . All resistors are of $1\text{ k}\Omega$ each. Opamp may be considered to be ideal . Your solution must be in full detail.



- (A) If V_{in} is connected to point C . systematically calculate V_{out} .
 (B) If V_{in} is connected to point A . Calculate current flowing through R_8 .
 (C) While V_{in} is connected to point A . Calculate V_{out} .
2. (A) Draw equivalent circuit diagram of a practical OPAMP.
 (B) What happens if OPAMP is used in open loop condition i.e. negative feedback is not employed. Explain in detail.
 (C) Explain the concept of negative feedback in operational amplifiers. How does negative feedback contribute to the performance and stability of op amp circuits? Provide a relevant example to illustrate your explanation.
3. (A) Design a non-inverting amplifier using an operational amplifier with a voltage gain of 10. Provide the necessary circuit diagram, along with the expressions for the input and output voltages. Assume ideal op amp characteristics and calculate the feedback resistor required for a desired gain.
 (B) Describe the operation of an inverting amplifier configuration using an operational amplifier. Derive an expression for the voltage gain of the inverting amplifier in terms of the input and feedback resistors. Provide a numerical example demonstrating the calculation of the gain for specific resistor values.

(C) Compare and contrast the characteristics and applications of ideal and non-ideal operational amplifiers. Discuss the effects of finite gain, bandwidth, input impedance, and output impedance on the performance of op amp circuits. Provide examples where non-ideal opamp behaviors significantly impact circuit design.

4. (A) Define and explain the following OPAMP parameters, Input offset voltage, total Output offset voltage, CMRR, slew rate, PSRR.

(B) Derive output voltage expression for differential amplifier using two OPAMPs. Also express the values of input impedances.

(C) Explain Instrumentation amplifier in detail under the following heads Circuit diagram, derivation of voltage gain, input impedances, advantages, uses and applications.

5. (A) Explain Integrator circuit both basic and practical circuit in which problems of basic circuit is eliminated while mentioning the problems with basic circuit and how can it be treated. Your answer must be in full detail.

(B) Explain differential input differential output amplifier and derive its input output relation.

(C) Explain Wien bridge oscillator under the following heads circuit diagram, condition of oscillation, frequency of oscillation, uses and area of application, limitations.

Institute of Engineering & Technology, DAVV Indore

B.E. II, Year (E&TC) A&B

Examination, April 2024

Subject Code – 4ETRC2, Subject Name – EMF&TL

Time: 3 Hr.

Maximum Marks: 60

Note: (1) There are five Questions in this paper, each Question has three sub parts a, b and c. Attempt any two parts from each Question and each Questions Carry equal marks.
(2) Provide Smith Chart graph paper.

- Q1 (a) Express the vector $\vec{F} = 2xyz \vec{a}_x - 5(x+y+z) \vec{a}_z$ in cylindrical coordinates. Find $|\vec{F}|$ at point P (2, 60°, 3) 6
- (b) Transform each of the following vectors to spherical coordinates at the point specified 6
- (i) $8\vec{a}_x$ at B ($r = 4, \theta = 25^\circ, \phi = 120^\circ$), (ii) $8\vec{a}_x$ at A ($x = 2, y = 3, z = -1$),
(iii) $5\vec{a}_x - 3\vec{a}_y - 5\vec{a}_z$ at P ($x = -2, y = -3, z = 4$)
- (c) (i) Determine the curl of the vector field $\vec{A} = \frac{\sin\theta}{r^2} \vec{a}_r - \frac{\cos\theta}{r^2} \vec{a}_\theta$ 6
- (ii) Determine the laplacian of the scalar field $U = \rho z \sin\theta + \rho^2 + z^2 \cos^2\theta$
- Q2 (a) A charge of $0.3 \mu\text{C}$ is located at point A (25, -30, 15) and a second charge of $0.5 \mu\text{C}$ at B (-10, 8, 12), Find $\vec{E}$ at (i) origin (ii) (15, 20, 50). 6
- (b) A potential field in free space is expressed as $V = \frac{60 \sin\theta}{r^2}$ volt. Find the electric flux density and hence the volume charge density at the point P ($r = 3, \theta = 60^\circ, \phi = 25^\circ$) in spherical coordinates. 6
- (c) The flux density $\vec{D} = \frac{r}{6} \vec{a}_r \text{ nC/m}^2$ is free space (i) find $\vec{E}$ at $r = 0.3 \text{ m}$ 6
- (ii) find the total electric flux leaving the sphere of $r = 0.3 \text{ m}$
- (iii) find the total charge within the sphere of $r = 0.4 \text{ m}$.
- Q3 (a) Find the vector magnetic field intensity in Cartesian coordinate at P (2, 4, 5) due to a Current filament of 10 A in $\vec{a}_z$ direction of the z axis extending from (i) $z = 0$ to $z = 8$ 6
- (b) A sheet of uniform charge $\rho_s = 5 \text{ nC/m}^2$ is present at plane $x = 4$ in free space and a line charge $\rho_l = 20 \text{ nC/m}$ is located at $x = 2, z = 5$. Find magnitude of electric field intensity at P (5, 6, 7) 6
- (c) In a material for which $\sigma = 5 \text{ S/m}$ and $\epsilon_r = 1$, the electric field intensity is $E = 250 \sin 10^{10} t \text{ V/m}$. Find the conduction and displacement current densities and the frequency at which both have equal magnitudes. 6

- Q4 (a) State and Prove Maxwell's equations and give their physical significance. 6
(b) State and explain Poynting theorem with respect to electromagnetic wave. 6
(c) Derive and explain Continuity equation for time varying fields. 6
- Q5 (a) Using Smith chart graphical representation method to find VSWR, Reflection Coefficient and input impedance of a lossless transmission line of electrical length 0.25λ is terminated with complex load impedance $Z_L = 125 + j125 \Omega$ and Characteristic impedance of 80Ω . 6
(b) A 85Ω lossless line connect a signal of 500 KHz to a load of 150Ω , if the load power is 65 mW . Determine (i) VSWR (ii) V_{min} and V_{max} (iii) Impedance at V_{min} and V_{max} . 6
(c) A lossless line having an air dielectric has a characteristic impedance of 400Ω . the line is operating at 200 MHz and $Z_{in} = 200 - j200 \Omega$, use Analytic method to find (i) VSWR (ii) Z_1 if the line is 1 m long (iii) the distance from the load to the nearest voltage maximum. 6

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B. E. II EXAMINATION APRIL 2024
Electronics & Telecommunication / Mechanical Engg.
4STRS4/4SMRS4 : Engineering Economics

[Max. Marks: 60]

Duration: 3 Hrs]

Note: Attempt any TWO parts from each question. Each carries 06 marks. Draw graph wherever necessary.

- Q.1 a) Differentiate between Micro and Macro Economics. (06)
 b) Write down Law of demand. Write exceptions to law of demand. (06)
 c) Explain Income and cross elasticity of demand with suitable examples. (06)
- Q.2 a) Write down different factors affecting supply. (06)
 b) Explain price Elasticity of supply. (06)
 c) Explain shift of the supply curve. (06)
- Q.3 a) Write short note on following: (06)
 i. Level of production
 ii. Scale of production
 b) Explain short run production function in detail. (06)
 c) Explain Break Even point. (06)
- Q.4 a) Explain Circular flow of income in simple economy. (06)
 b) Write Value added method for national income calculation. (06)
 c) What is personal income and personal disposable income? (06)
- Q.5 a) Describe fiscal policy and its types. (06)
 b) Write down tools of monetary policy in detail. (06)
 c) Explain goals of Monetary policy. (06)

B.E. II EXAMINATION APRIL- 2024
E & TC ENGG.

4ETR4C1: MEASUREMENT SYSTEM AND SENSOR

Duration: 3 Hrs.

Max. Marks : 60

Note: Attempt any two parts of each question. Each question carries equal marks. Make suitable assumptions if necessary.

- Q.1** (a) Explain each block of Generalized Measuring System. 6
 (b) (i) Explain Systematic error. 3 + 3
 (ii) A temperature transducer has a range of 0 to 100°C and an accuracy of $\pm 0.5\%$ of FSV. Find the error in a reading of 55°C.
 (c) (i) Explain the theory of resistive strain gauge. 3 + 3
 (ii) A strain gauge circuit uses a resistance of 120Ω and $F=2$. When working under maximum sensitivity conditions the current flow through the system is limited to 20mA. Calculate the supply voltage and change in output voltage when a strain of $500\mu\text{strain}$ is impressed on the gauge.
- Q.2** (a) Explain the working of Thermocouple along with its laws. 3 + 3
 (b) (i) Discuss the working of Inclined Manometer. 3 + 3
 (ii) A single column manometer is using mercury of specific gravity 13.6 as the manometric liquid. To what height will the mercury rise in the narrow limb if a differential pressure of 75 KN/m^2 is applied. The wide and the narrow limb diameters are 160 mm and 6 mm respectively.
 (c) (i) Explain the working of Optical Pyrometer. 3 + 3
 (ii) A Platinum resistance thermometer has a resistance of 150Ω at 100°C . When exposed to high temperature gas, the resistance increases to 300Ω . Determine the temperature of hot gas. It may be presumed that at 0°C , the resistance is 100Ω and that the temperature coefficient of resistance for platinum is $0.00390^\circ\text{C}^{-1}$.
- Q.3** (a) (i) Explain the working of LVDT. 3 + 3
 (ii) A quartz crystal having a thickness of 2mm and a voltage sensitivity of $0.05\text{V}\cdot\text{m}/\text{N}$ is subjected to a pressure of $15 \times 10^5 \text{ N/m}^2$. Calculate the voltage developed by the piezo-electric pick up. If permittivity of quartz is $40.6 \times 10^{-12} \text{ F/m}$, calculate its charge sensitivity.
 (b) How linear and angular displacement is measured with Capacitive gauge. 3 + 3
 (c) Discuss the working of (i) Drag cup tachometer (ii) Photoelectric tachometer. 3 + 3
- Q.4** (a) (i) Define Dry bulb and Wet bulb temperature. 6
 (ii) How humidity is measured with Hair Hygrometer.
 (b) How discharge is measured with (i) Rectangular and (ii) Triangular Weir? 3 + 3
 (c) How level is measured with (i) Bubbler tube and (ii) Resistive gauge. 3 + 3

- Q. 5**
- (a) (i) What is the use of Pitot tube? Explain in detail. 3 + 3
(ii) A submarine moves horizontal in sea & its axis is much below the surface of water. A pitot tube just placed in front of the submarine & along its axis, is connected to the limbs of U-tube containing mercury; a 17cm difference in the mercury level is observed. Find the speed of submarine knowing that density of mercury is 13.6 and that of sea water is 1.026 with respect to fresh water.
 - (b) Discuss, with proper sketch, the working of CCD device. 6
 - (c) Explain (i) Semiconductor gas sensor (ii) NDIR gas sensor. 3 + 3

Roll no.....

BE 2nd year. Final examinations, April 2024
Electronics & Telecommunication (RE)
4ETRG2 DIGITAL SIGNAL PROCESSING

Time: 3 hours

Max marks: 60

Note: Solve any two parts from each question. All questions are compulsory and carry equal marks.
 Make suitable assumptions whenever required.

- Q.1) (a) (I) Determine whether the discrete time system $y(n) = \cos(\omega n)$ is
 i) Memory less ii) Stable iii) Causal iv) Linear v) Time invariant vi) Invertible
 (II) Write the condition for complex exponential sequence $x(n) = e^{j\omega n}$ to be periodic.
 (b) Explain Linear Time variant and Linear time invariant system with example.
 (c) What do mean by Energy signal and Power signal in discrete time system,
 write two examples of both signals. Also determine whether $e^{j\omega n} u[n]$ is energy
 signal or Power signal

- Q.2) (a) State and prove Time scaling and Time shifting properties in Z transform.
 (b) (I) Determine $x[n]$ for the following $X[z] = z^{-2}(1+0.3z^{-1}-0.7z^{-2})(1+z^{-3})$,
 where $X[z]$ is z transform of $x[n]$.
 (II) Write properties of ROCs (Region of convergence)
 (c) Calculate Z transform of $\cos\omega_0 n u[n]$ and $\sin\omega_0 n u[n]$

- Q.3) (a) What do you mean by Causality and Stability of system. Also explain with example
 how to determine in Z transform, if system transfer function is given.
 (b) Using Long Division method, determine the Inverse Z transform of

$$X[z] = \frac{z+2}{2z^2-7z+3}$$

If ROCs are (I) $|z| > 3$ (II) $|z| < 1/2$ (III) $1/2 < |z| < 3$

- (c) Determine the response of following discrete system:
 $x(n+2) - 3x(n+1) + 2x(n) = \delta(n)$, where $\delta(n)$ discrete impulse signal
 Assume that all initial conditions are zero

- Q.4) (a) State and prove Time and Frequency shifting property in DTFT
 (b) Find the stability and frequency response of DT-LTI system
 represented by the impulse response: $h(n) = 1/8 (7/8)^n u(n)$
 (c) Compute the DFT of the following sequences
 (I) $[1, 1, -1, 1]$ (II) $[j, -j, j, -j]$

- Q.5) (a) Obtain Cascade form realization of
 $y(n) - (1/4)y(n-1) - (1/8)y(n-2) = x(n) + 3x(n-1) + 2x(n-2)$
 (b) Determine and sketch Direct Form I and Direct Form II realizations for following
 pulse transfer function. $H(z) = (1+2z^{-1}+0.7z^{-2}+2z^{-3}) / (2-3z^{-1}-4z^{-2}-0.8z^{-3})$
 (c) What do mean by sampling in Discrete Time System. State and prove sampling theorem.